

Problem

In a TV transmitter, a series circuit has an impedance of $3\text{ k}\Omega$ and a total current of 50 mA . If the voltage across the resistor is 80 V , what is the power factor of the circuit?

Step-by-step solution

Step 1 of 1

$$Z = R \pm jX$$

$$V_R = IR \Rightarrow R = \frac{V_R}{I} = \frac{80}{50 \times 10^{-3}} = 1.6\text{ k}\Omega$$

$$\begin{aligned} |Z|^2 &= R^2 + X^2 \Rightarrow X^2 = |Z|^2 - R^2 \\ &= 3^2 - (1.6)^2 \end{aligned}$$

$$X = 2.5377\text{ k}\Omega$$

$$\theta = \tan^{-1}\left(\frac{X}{R}\right)$$

$$= \tan^{-1}\left(\frac{2.5377}{1.6}\right)$$

$$= 57.77^\circ$$

$$P_f = \cos \theta = 0.5333$$